HSLU Lucerne University of Applied Sciences and Arts

DATA VISUALIZATION FOR AI AND ML

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PROJECT DESCRIPTION

1. Administrative

The idea of a project is to apply the foundations we have discussed in the course in a work of your own.

There are two projects required to pass the class:

- a mini-project
- a final project

A mini-project is done individually. In this project, you will reproduce one data visualization from a list of visualizations I will provide.

For the final project, you should adhere to the following guidelines:

- You can work alone or in a team of 2 or 3 people
- You choose one of the 2 project types
- You present the project in the final week of the course (ungraded)
- You have at least one coaching session with me about your project (ungraded)

The final project substitutes for a final exam. This means that in the project and the corresponding report you have to convince me that you understood and applied the material discussed in the course.

2. Important dates

30.09.2024: list of visualizations for the mini-project is available on ILIAS

28.10.2024: submit the mini-project to ILIAS

04.11.2024: enter the names of group members and the project title in ILIAS

19.11-10.12.2024: coaching sessions per zoom

17.12.2024: project presentations and peer feedback session

07.01.2025: deadline for final submissions *(before the exam session)*

3. Mini-project

In the mini-project, you will pick one data visualization from a list and you will reproduce it with Matplotlib and Matplotlib-based libraries (such as Seaborn). The list will be available on ILIAS.

As a submission, I expect a single Jupyter notebook containing the code used to reproduce a picked visualization. You should aim to be as close as possible, but some deviations from the original are acceptable (e.g. using a similar font but not the exact same one, positioning of annotations can be close enough). You should try to make the code clean and modular (that is, use variables and functions when required), but I don't expect you to develop your own classes or modules. You should follow the guidelines for using PLT API and OO API of Matplotlib as discussed in the class.

The project is done individually. I expect a workload of 6-10 hours. Throughout the course I will give you many examples of "how to do X in Matplotlib" - I would like you to practice them in the mini-project.

4. Final project types

4.1. Data storytelling

You submit a data story using one or more datasets of your choice. Visualizations may be static or interactive. I want you to focus on storytelling and design, intentional use of annotations and other elements we discussed. You need to include at least 4 charts which constitute a logical story (the minimum requirement increases by +1 chart for each additional team member), bonus points if you create custom charts or heavily customize the pre-built charts. You also need to include text that presents the actual story and guides the reader through the conclusions. It can be a static html or an interactive site with controls and animations (e.g. forward or back). Please do not submit a pdf.

Inspiration:
https://pudding.cool
https://fivethirtyeight.com
https://distill.pub
https://explorabl.es
https://www.srf.ch/news/srf-data
https://www.nzz.ch/visuals
https://www.economist.com/graphic-detail
https://graphics.reuters.com

I do not expect you to deliver such a complex project as those listed as inspiration. Each of the articles listed above needed much more work than you can possibly spend on the project. Get inspired, but pick something much smaller, use a dataset that is interesting to you and have fun while learning how to do it.

4.2. Visual analytics

You create a tool that helps with a particular analytic task or lets the user explore the results of your work or interact with your model/analysis (e.g. simple classification model, clustering, dimensionality reduction). The task and target user need to be defined and described in the report. There need to be at least 2 interactive charts with filtering elements and interactivity between them (the minimum requirement increases by +1 chart for each additional team member). I want you to focus on providing interactivity that supports the task and on choosing chart types that support the target user.

Inspiration:
https://explorabl.es

https://distill.pub
https://streamlit.io/gallery
https://dash.gallery/Portal
https://research.google/teams/brain/pair
https://poloclub.github.io

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5. Final project submission

The deliverables for the final project constitute of 3 parts: the project, the report and the work summary. Please pack them all into one file (zip, rar etc.) and upload it in ILIAS.

5.1. Project

As project outcomes, I would like you to submit:

- A short video (max 30s) where you demo your work
- Source code used to create the data story / app with the requirements.txt/pyproject.toml files and a readme (and any other information I need to run the code locally)
- The actual data story / app (e.g. the compiled html if applicable)

In your code, use human-friendly names of variables and functions and provide a brief commentary of what parts of the code do.

5.2. Report

Length of the report: 7500-11000 characters (with spaces, but w/o TOC and work table) - around 4-6 A4 pages. If the report is significantly shorter or longer than this, it will not get the full grade.

The report needs to be written in English. I won't grade the grammar, but I will not give the full score if there are very visible and frequent spelling mistakes and typos.

What should a report contain:

- your motivation for picking up the topic
- description of the exploratory work you did (data exploration, a review of similar work that has been already done with citations - these don't have to be peer-reviewed papers, but can also be web-apps, journal articles etc.)
- short description of the target audience of your visualizations and how it influenced your choices
- explanation why you chose particular chart types and other visualization elements like colors, annotations, insets, styling
- list of libraries and packages you used with a short motivation why you used those and not the alternatives
- citations for the data source and other relevant sources

If you did an extensive data exploration and prototyping, you can submit the notebooks, sketches or other artifacts as a separate attachment to the report. They will not be graded as separate items, but will help me understand your choices and may work towards a better grade.

5.3. Work summary

A table summarizing who did when and how long what task during the project phase, for example:

Date	Hours	Done by	Task description
01.05.2022	1	XY	Set up git repository and working environment

For every team member, please provide the total number of hours as well.
I expect the total project workload to be around 35-50 hours per person.

6. Grading criteria

The grade is composed as following:

10% mini-project

50% final project:
- motivation and storytelling
- complexity and completeness
- use of existing tools or libraries vs custom work
- application of what we discussed in the lecture
- all the artifacts are present (topic on time, submitted code, exports, requirements for environment…)
- project runs locally
- *if using ML: model accuracy and model complexity has no influence on the grade*

40% report for the final project:
- description of the process
- exploratory work
- motivation behind the choices you made
- logical flow
- quality (no typos etc.), language appropriateness
- *report needs to be tidy and complete, extra graphical design elements such as a picture on the title page will have no influence on the grade*
- *small grammar mistakes or language simplicity will not influence the grade as long as you use correct terms we discussed in the lecture and make yourself clear*

I will provide detailed project feedback on request only.

7. Possible data sources (a not exhaustive list)

https://www.kaggle.com/datasets
https://opendata.swiss
https://data.stadt-zuerich.ch (and other cities / kantons)
https://www.bfs.admin.ch/bfs/de/home.html
https://www.makeovermonday.co.uk
https://www.tidytuesday.com
https://data.fivethirtyeight.com
https://ourworldindata.org

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8. Some tips for a successful final project

You only have a chance to get the best grade if you submit both the mini-project and the final project.

When defining your project topic, pick a good dataset right away (not too simple, not too complex), to avoid switching the dataset later in the semester.

The project may be related to another project that you did for a different class. In that case, the other project that you base on cannot have had a data visualization component to it. You should also disclose in the report which part of the project was done for the other class, so that there is a clear cut.

It is possible to use another tool than we discussed in the class (e.g. if you are a front-end developer and want to do a JS-based app), as long as you can convince me in the report it is a good choice for what you want to achieve.

I am not expecting you to make a scientific discovery or to uncover a journalistic story. I am only grading how well you can achieve your communication goals using data visualization and the concepts we discussed in the class.

Spend some time looking at how other people have visualized a similar dataset or a similar message (blogs, journals etc.). If you get inspired, disclose the inspiration in the report. Some of your charts can be similar to the inspiration, but you need to make them distinct enough to call them your own.

There should be a clear distinction between the project and the report. In the project, you describe your insights, while in the report you describe your process. Also, do not waste your report character count to describe the insights and discoveries you made during the data exploration. If you want to share them with me, attach a ipynb notebook instead. In the report I want you to describe what you wanted to communicate and how you decided which visual means and dataviz concepts were needed to achieve it.

Double-check the list of deliverables to make sure your submission is complete. Don't forget the recording. Make sure that the virtual environment and the project run by themselves - I need to be able to run the project locally to give it a good grade.

The time you can spend on the project is limited. You will have to prioritize. If in doubt about time priorities, it's better to code something simpler but customize the chart more (e.g. work on design, annotations, colors, storytelling rather than produce a multitude of simple charts generated with default styling), and spend more time on making the right choices and describing your process. It pays off to start working on the project early in the semester.

Use the project presentation during the last class to get and give valuable feedback from your peers. For example: do the visualizations come across are you intended? is there a design or coding struggle that they could help you with? When giving feedback, share your thoughts in a constructive way, be friendly and helpful.

I try to finish most of the lectures a bit early in case of questions. When in doubt about something related to the project, come and ask, send me an email, or make use of the course forum and peer knowledge.

If you are happy with how the project turned out, I encourage you to publish it online as a part of your portfolio!

Good luck!

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